

Total No. of Questions : 4+4]

[Total No. of Printed Pages : 8

Roll No. _____

REF NO. 058544 & 058843

B. Sc. (Hons) Semester IV Examination

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CHEMISTRY

**Paper No. CHB-401 : Inorganic Chemistry–II
and Organic Chemistry–III**

Time : 3 hours

Full Marks : 70

(Write your Roll No. at the top immediately on the receipt of this question paper)

The figures in the right-hand margin indicate marks.

USE SEPARATE ANSWER BOOK FOR EACH SECTION

**Section—A
(Inorganic Chemistry–II)
(Marks : 35)**

*Answer **three** questions including Question 1 which is compulsory*

1. 1. Arrange the following acids in decreasing order of acidic strength in (i) H₂O and (ii) CH₃COOH as solvent. Justify your answer:
HF, HCl, HClO₂

3
2. 2. What are the differences in characteristic properties of blue colored solution and bronze colored phase obtained by addition of alkali metals in liquid ammonia? **3**
3. 3. What are the limitations of effective atomic number rule? **2**
4. 4. The water solubility of lithium halides is in the order of LiBr > LiCl > LiF. However, this trend is reversed in case of analogous silver halides. Explain the reason behind it. **3**
2. 1. Give two examples each of protic and aprotic solvents and write autoionization reaction for them. **3**
2. 2. Predict the product when (CH₃)₂N-PF₂ is reacted with (i) BH₃ and (ii) BF₃. Explain the reactivity difference in the two cases. **3**
3. 3. State the properties that determine the utility of a solvent. **3**
4. 4. What is lanthanide contraction and what are its important consequences? **3**
3. 1. Write one example each of the following types of octahedral complexes. Draw the various stereoisomers and specify the enantiomeric pairs in each case : **3**

- (i) Ma_3b_3
- (ii) $\text{M}(\text{AA})_2\text{b}_2$

6

2. Write the IUPAC names of the following :

- (i) $\text{K}_2[\text{Zn}(\text{NCS})_4]$
- (ii) $[\text{Mn}(\text{acac})_3]$
- (iii) $[\text{Pt}(\text{NH}_3)_4][\text{PtCl}_4]$

4

(iv) Render structural formula layout for the bridged carbonyl system:

μ -CO bridged carbonyl coordination framework of $\text{Fe}_2(\text{CO})_9$

3. Why is $[\text{Ni}(\text{en})_3]^{2+}$ more stable than $[\text{Ni}(\text{NH}_3)_6]^{2+}$ though both are octahedral? Draw their structures. 2

4. 1. Using Valence Bond theory predict the structure and spin only magnetic moment of $[\text{Ni}(\text{CO})_4]$, $[\text{NiCl}_4]^{2-}$, $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{Ni}(\text{NH}_3)_6]^{2+}$. 4

2. Write the electronic configuration and number of unpaired electrons in the following :

Ce, Gd, Eu, Tb

4

3. Predict the feasibility of the following reactions in water and liquid ammonia along with the reason:

- (i) $\text{BaCl}_2 + 2\text{AgNO}_3 \longrightarrow 2\text{AgCl} + \text{Ba}(\text{NO}_3)_2$
- (ii) $\text{AgCl} + \text{KNO}_3 \longrightarrow \text{KCl} + \text{AgNO}_3$

2

4. What is the difference between ammoniation and ammonolysis? Explain with one example of both. 2

Section—B
(Organic Chemistry—III)
(Marks : 35)

*Answer **three** questions including Question 1 which is compulsory*

- 1. 1. Write the chemical structure of α and β D(+) glucose. 2
- 2. Discuss the synthesis of the following compounds with mechanisms : 2+2
 - (i) Toluene $\longrightarrow$ 2-Methylnaphthalene sequence
 - (ii) Diethyl malonate derived β -ketoester conversion $\longrightarrow$ Pyrrole derivative framework
- 3. Discuss the orientation of electrophilic substitution reaction in pyridine. 2

4. Write the structure of the following : $1\frac{1}{2} + 1\frac{1}{2}$
- (i) Fluorescence
 - (ii) Malachite green.
2. 1. Write the stepwise mechanisms of the following reactions : $2\frac{1}{2} + 2\frac{1}{2}$
- (i) Kiliani-Fischer synthesis
 - (ii) Wohl's degradation method.
2. Explain the reason(s) of the following : **2+2**
- (i) Anthracene is very reactive at 9 and 10 positions.
 - (ii) Pyridine N-oxide is more electrophilic than pyridine.
3. How will you establish the structure of indigotin? Explain. **3**
3. 1. Write the product(s) in the following reactions : **6×1**
- (i) 2-Furaldehyde $\xrightarrow[\text{(ii) [O]}]{\text{(i) KCN}}$ **P**
 - (ii) 1-Naphthol $\xrightarrow{\text{HNO}_2}$ **M + N**
 - (iii) Pyridine $\xrightarrow[320\text{ }^\circ\text{C, H}_3\text{O}^+]{\text{KOH/ air}}$ **R**
 - (iv) 1-Tetralone $\xrightarrow{220\text{ }^\circ\text{C}}$ NaOH-KOH (1:1) **Q**
 - (v) Phenanthrene + Maleic anhydride $\xrightarrow{\Delta}$ **Z**
 - (vi) Naphthalene $\xrightarrow[\text{(ii) Zn, H}_2\text{O}]{\text{(i) O}_3}$ **R**
2. How will you prepare the following compounds from D(+) glucose? Write down the stepwise mechanisms : **2+2**
- (i) Fructose
 - (ii) Mannose
3. Discuss the mutarotation in brief. **2**
4. 1. How do epimers and anomers differ? **2**
2. Complete the following reactions : **5×1**
- (i) Thiophene $\xrightarrow{\text{Raney Ni / H}_2}$?
 - (ii) Anthracene $\xrightarrow{\text{Aq. HNO}_3}$?
 - (iii) 2 Benzene + 2CH₂Br₂ $\xrightarrow[\text{(ii) Pd-C}]{\text{(i) AlCl}_3}$?
 - (iv) D-Erythrose + $\begin{matrix} \text{H}_2\text{C-NO}_2 \\ \text{H}_2\text{C-OH} \end{matrix}$ $\xrightarrow[\text{(ii) H}_2\text{SO}_4]{\text{(i) CH}_3\text{ONa}}$?
 - (v) Indigotin $\xrightarrow{\text{HNO}_3/\text{[O]}}$?

3. Write the synthesis of indigotin from anthranilic acid with mechanistic details. **3**
4. Why do malachite green and fluorescein develop colour in solution? Explain. **2**

ExternalPortals&AcademicResources:

- **ExaminationPortal:** <https://bhu-exam.vercel.app/>
- **StudentResourceCenter:** <https://quashan-me.vercel.app/>
- **AcademicPortfolio:** <https://me-aryan.vercel.app>